

Vera L. Casquero

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EDUCATION

University of Notre Dame – College of Engineering – Bachelor of Science in Engineering **Notre Dame, IN**
Major: Computer Science | **Minor:** Engineering Corporate Practice **May 2027**
Coursework: Data Structures, Systems Programming, Discrete Math, Linear Algebra & Differential Equations, Integrated Engineering & Business Fundamentals

RELEVANT WORK EXPERIENCE

Iris Recognition Research Internship **Notre Dame, Indiana**
Intern *May -December 2024*

- Led the development of a **Python-based system** to classify and filter iris data, increasing dataset accuracy by up to **25%**, enhancing research efficiency and data reliability.
- Co-engineered a **Python** algorithm that automated image reporting and data processing, reducing manual labor by **50+** hours, demonstrating process optimization and operational efficiency.
- Employed **machine learning (ResNet-18 architecture)** to evaluate a system performance, achieving **100% accuracy** in distinguishing authentic irises images from attack images.

ACTIVITIES & PROJECTS

Student International Business Council **Boston, MA**
Amazon – Project Leader *January 2025 – Present*

- Leading a cross-functional team of **10+** students to develop strategic enhancements for Amazon Prime's **UX**, aligning technical solutions with key business objectives.
- Serving as the primary liaison with Amazon representatives, strengthening client communication and delivering data-driven recommendations that align with user needs and market trends.

Amazon – Travel Team Analyst *October 2024 – December 2024*

- Conducted market research and data analysis, projecting a **10-15%** increase in customer retention rates through proposed interface improvements.
- Delivered a data driven proposal to optimize Amazon's homepage and improve user engagement using **Generative AI**, integrating an **RCNN** framework for personalized recommendations, driving higher sales conversions.

Post-graduate Employment App **Notre Dame, Indiana**
Developer *Abril 2024 – May 2024*

- Developed a data-driven app analyzing employment outcome for graduates across demographics focused on STEM fields, serving over **2,000+ data points** utilizing **MATLAB** frameworks.
- Created an algorithm to optimize data processing and analysis by up to **30%**, delivering faster and clear insights on how users' degrees and demographics influence employment status.

LEADERSHIP & SERVICE

Fundación Mentres Brillantes **Santo Domingo, Dominican Republic**
Founder/President *May 2022 – Present*

- Established and led the first organization dedicated to teaching programming and STEM subjects to children in Dominican Republic's public school.
- Led fundraising initiatives that generated **\$50,000** in support for five public schools, directly addressing the digital divide and providing over **1,000 students** with upgraded educational resources.
- Launched a tech education program currently benefiting **65 students**, now scaling it to reach **300+ students** next year.

Notre Dame Computer Club **Notre Dame, Indiana**
Secretary *January 2025 – Present*

- Boosted club engagement by **10%** through dynamic initiatives, including speaker forums and programming workshops.
- Leading the development of a competitive hackathon team to represent the club in various programming challenges, enhancing members' problem solving, teamwork, and technical skills.

RELEVANT SKILLS

Technical: Python, C/C++, SQL, Linux, AlbaCore, SolidWorks, Microsoft Excel, MATLAB, Figma

Societies: Association of Latinos Professionals for America, Society of Women Engineers, Latino Honor Society

Language: Spanish (Fluent), English (Fluent)

Interests: Golf | Reading (Auto-biographies) | Traveling | International Cuisine